



MEDICAL DIAGNOSIS EXPERT SYSTEM

*A Desktop Application for Disease Diagnosis Using
Rule-Based Inference and Weighted Scoring*



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1. Project Overview

Project Title

Medical Symptom Decision System

Project Type

Desktop Application using Java Swing and Microsoft SQL Server.

Project Description

The Medical Symptom Decision System is a desktop-based healthcare decision support application developed using Java and Microsoft SQL Server. The system assists users in analyzing symptoms and generating possible disease predictions based on a weighted scoring algorithm.

The application simulates a simplified medical expert system by connecting symptoms, diseases, and weighted medical relationships stored dynamically in a database. The system is capable of managing patients, storing diagnosis history, generating statistics, and dynamically updating diseases and symptoms through an admin panel.

The project follows a modular architecture inspired by the MVC (Model-View-Controller) design pattern to ensure scalability, maintainability, and clean code organization.

2. Project Objectives

The main objectives of the project are:

- Build a healthcare decision-support system.
 - Use database-driven logic instead of hardcoded rules.
 - Implement real-time diagnosis and patient history tracking.
 - Create a scalable architecture for future expansion.
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3. Technologies Used

Technology	Purpose
Java	Core programming language
Java Swing	Graphical User Interface
Microsoft SQL Server	Database Management

4. System Features

4.1 Diagnosis System

- Select symptoms dynamically from database.
- Analyze symptoms using weighted scoring algorithm.
- Display top possible diseases.
- Show confidence percentages.
- Color-coded severity levels.

4.2 Patient Management

- Add new patients.
- Delete patients.
- Store patient information.
- Connect diagnoses to specific patients.

4.3 History Management

- Store diagnosis results.
- View patient diagnosis history.
- Filter history by patient.
- Sort diagnosis records by date.

4.4 Statistics Dashboard

- Total patients count.
- Total diagnoses count.
- Most common diseases.
- Real-time statistical information.

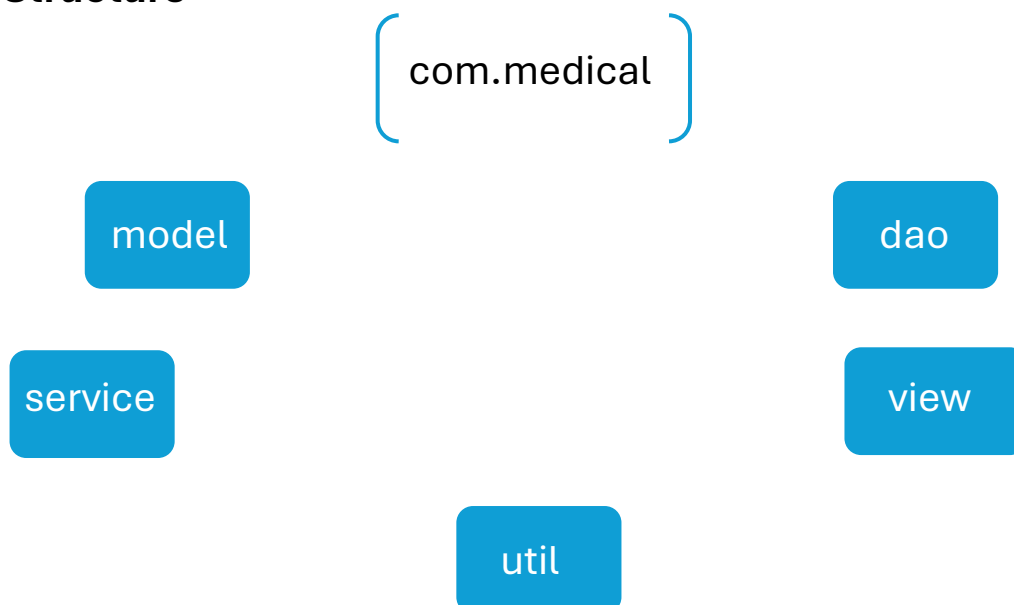
4.5 Admin Panel

- Add new symptoms.
- Add new diseases.
- Link diseases with symptoms.
- Set disease-symptom weights.
- Dynamically expand system knowledge.
- Control patients existence and data in system

5. System Architecture

The system is organized using layered architecture.

Package Structure



Package Responsibilities

model

Contains all data classes:

- Patient
- Disease
- Symptom
- Diagnosis
- SymptomInput

dao

Responsible for database operations:

- Insert
- Delete
- Update
- Retrieve

service

Contains business logic and decision engine.

view

Contains Java Swing GUI components.

util

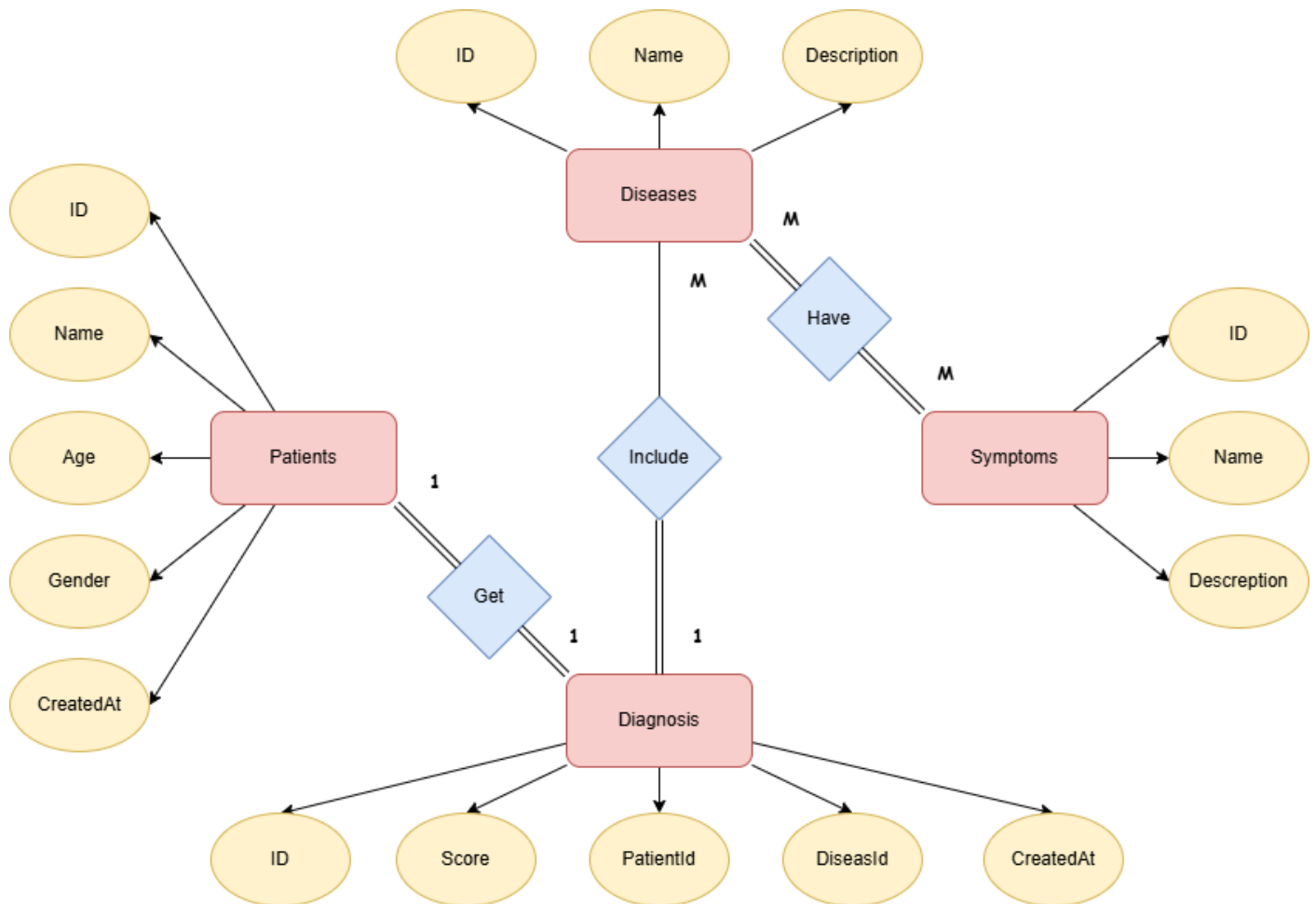
Contains utility classes such as database connection.

6. Database Design

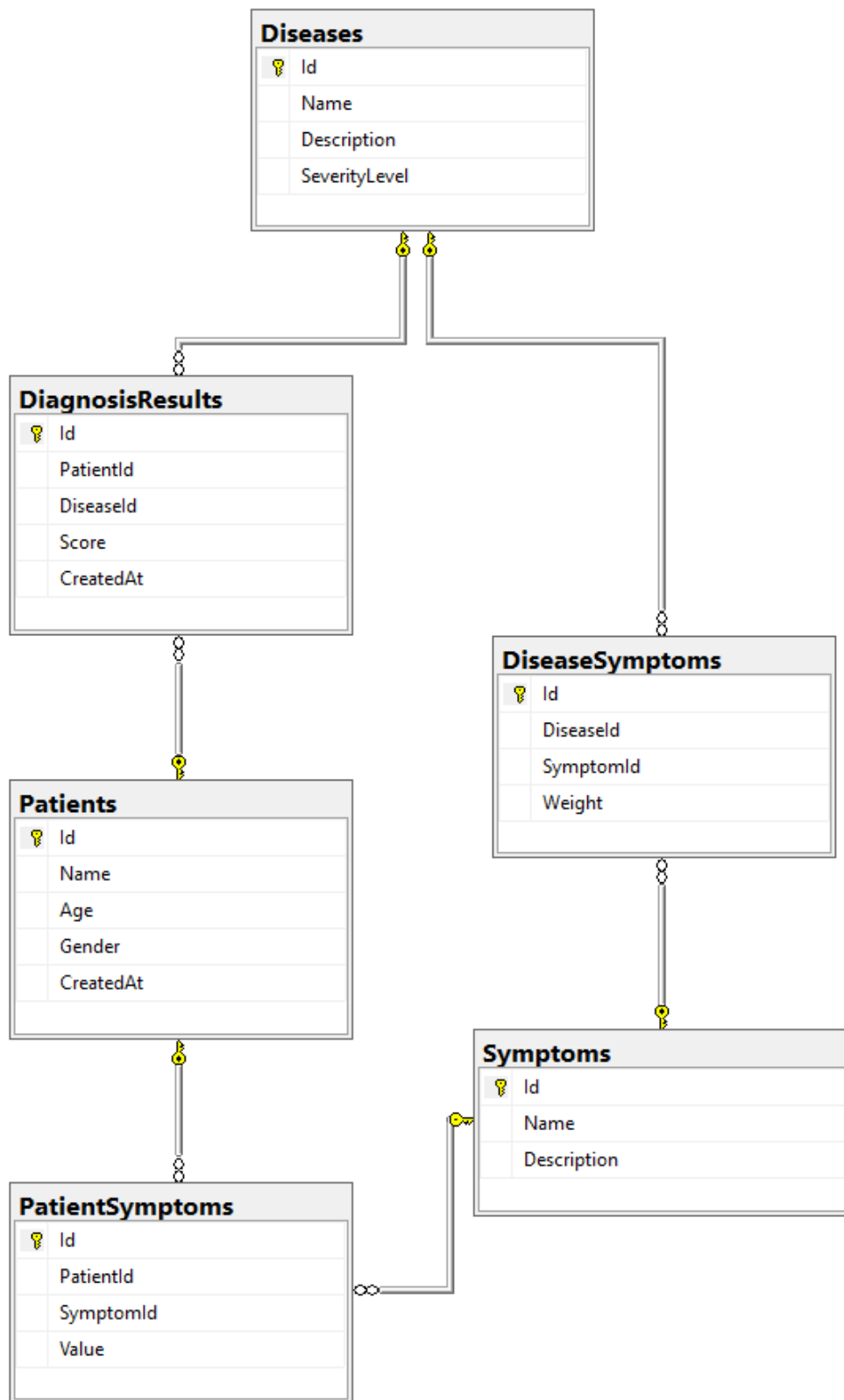
Database Name

Medical_Symptom_Decision

ERD



Mapping



7. Decision Engine Logic

The core logic of the project is based on a weighted scoring system.

Formula

$$\text{Score} = \sum(\text{weight} \times \text{symptom existence})$$

Where:

- Weight = importance of symptom for disease.
- Symptom existence = 1 if selected, 0 otherwise.

Example

If:

- Fever weight = 3
- Cough weight = 2
- Fatigue weight = 3

Then:

$$\text{Flu Score} = 3 + 2 + 3 = 8$$

The diseases are sorted by highest score to determine the most probable conditions.

8. GUI Design

The application uses Java Swing with a tabbed interface.

Main Tabs

Diagnosis Tab

- Symptom selection panel
- Analyze button
- Result cards
- Confidence percentages

The screenshot shows a Java Swing window titled "Medical Decision System" with a standard Mac OS X title bar (yellow background, red, yellow, and green buttons). The window contains a tabbed interface with five tabs: "Diagnosis" (selected), "Patients", "History", "Statistics", and "Admin".

Under the "Diagnosis" tab, there is a "Patient" section with a "Select Patient:" label, a dropdown menu showing "Ahmed (ID: 1)", and a blue "Refresh" button. Below this is a "Select Symptoms" section with a list of 15 symptoms, each preceded by an unchecked checkbox:

- Fever
- Cough
- Fatigue
- Loss of Appetite
- Headache
- Nausea
- Vomiting
- Diarrhea
- Joint Pain
- Swelling
- Shortness of Breath
- Chest Pain
- Dizziness
- Sore Throat
- Runny Nose
- High Temperature

At the bottom of the "Select Symptoms" panel is a blue "Analyze" button. The right side of the window is a large, empty light gray area, likely for displaying results.

Patients Tab

- Add patients
- Delete patients
- View patient list

The screenshot shows a web application window titled "Medical Decision System". It features a navigation bar with five tabs: "Diagnosis", "Patients" (which is the active tab), "History", "Statistics", and "Admin". The main content area is divided into two columns. The left column contains three labels: "Name:", "Age:", and "Gender:". The right column contains three corresponding input fields: a text box for the name, a text box for the age, and a dropdown menu for the gender. The dropdown menu is currently set to "Male". Below these input fields, there is a large blue button with the text "Save Patient".

Medical Decision System

Diagnosis Patients History Statistics Admin

Name:

Age:

Gender: Male

Save Patient

History Tab

- View diagnosis records
- Filter by patient

Medical Decision System

Diagnosis

Patients

History

Statistics

Admin

Filter History

Select Patient:

David (ID: 4)

Load History

Refresh

Patient	Disease	Score	Date
David	Flu	2.0	2026-05-07 22:09:01.357
David	Migraine	3.0	2026-05-07 22:09:01.34

Statistics Tab

- Display system analytics

Medical Decision System

Diagnosis

Patients

History

Statistics

Admin

Total Patients: 6

Total Diagnoses: 7

Top Diseases:

- Food Poisoning (2)
- Migraine (2)
- Flu (1)

Refresh Stats

Admin Tab

- Add diseases

The screenshot shows a web application window titled "Medical Decision System". It has a top navigation bar with tabs: "Diagnosis", "Patients", "History", "Statistics", and "Admin". The "Admin" tab is selected. Below the navigation bar, there are four sub-tabs: "Add Symptom", "Add Disease", "Link Disease", and "Manage Patients". The "Add Disease" sub-tab is active. The main content area contains a form with the following fields:

- Disease Name:** A text input field.
- Description:** A text input field.
- Severity:** A dropdown menu currently showing "Low".
- Add Disease:** A blue button with a gradient effect.

- Add symptoms

The screenshot shows the same "Medical Decision System" window, but with the "Add Symptom" sub-tab selected. The form fields are:

- Symptom Name:** A text input field.
- Description:** A text input field.
- Add Symptom:** A blue button with a gradient effect.

- Link disease relationships

Medical Decision System

Diagnosis Patients History Statistics Admin

Add Symptom Add Disease Link Disease Manage Patients

Disease: Common Cold

Symptom: Fever

Weight (1-5):

Link

- Manage Patients

Medical Decision System

Diagnosis Patients History Statistics Admin

Add Symptom Add Disease Link Disease Manage Patients

Name Age Gender

Male

ID	Name	Age	Gender
1	Ahmed	25	Male
2	Sara	30	Female
3	Omar	40	Male
4	David	20	Male
6	Johnathan	25	Male
7	Alia	30	Female

Delete Selected

9. System Workflow

1. User selects a patient.
 2. User chooses symptoms.
 3. Decision Engine processes symptoms.
 4. System calculates disease scores.
 5. Top diseases are displayed.
 6. Diagnosis is stored in database.
 7. History and statistics are updated.
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10. Future Improvements

Potential future enhancements include:

- AI-based diagnosis prediction.
- Machine learning integration.
- Dark mode UI.
- Real-time charts and graphs.
- PDF report export.
- Web-based version.
- Multi-user authentication.
- Cloud database integration.
- Search engine for symptoms.
- Advanced filtering and analytics.